3 Dimensional Category Learning

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Motivation

We initially sought to analyze an existing broad dataset with one of the models we explored in our course. However, after only finding mediocre datasets and consulting our instructor, we pivoted to extending an existing model. Nosofsky's *Attention, Similarity, and the Identification-Categorization Relationship* then became the main focus of our project, as we investigated how changing the number of varying dimensions in Nosofsky's experiment would affect the model's performance and impact how biases and weights were assigned to the model's fit of human data. Ultimately, we hoped to investigate the impact that increasing the number of dimensions considered when learning categories would have on how the categories were learned (by means of the model's parameters and subject data). Before we began exploring this topic, we searched for research extending Nosofsky's model and found minimal work on this topic, which motivated us to explore it further. Humans often learn categories in more than just two dimensions, (i.e. choosing a fruit that is ripe based on smell, color, and price), so our expansion of this experiment has some real-world bearing. How one weighs these dimensional characteristics or how one is biased toward picking a certain category can be analyzed by Nosofsky's generalized context model (GCM).

Methods

The original model Nosofsky developed showed how subjects identify and categorize a set of confusable stimuli that each vary on two given dimensions. The

probability of some stimulus S_i with parameters varying on two distinct dimensions being a part of category R_1 is defined as:

$$P(R_1|S_i) = \frac{b_1 \sum_{j \epsilon C_1} \eta_{ij}}{b_1 \sum_{j \epsilon C_1} \eta_{ij} + (1-b_1) \sum_{k \epsilon C_2} \eta_{ik}}$$

Where b_1 is the bias parameter for Category 1. Since there are only two categories, the expression can be written in this way. The η functions are defined by the difference in similarity between two stimuli (i and j):

$$\eta_{ij} = e^{-(c[\sum_{k=1}^{N} w_k |x_{ik} - x_{jk}|^r]^{1/r})^r}$$

The x_{ik} values are given by the multi-dimensional scaling solution for the stimulus set (we call these features in our Homework 3 modification), the c parameter is a scaling parameter we adjusted to fit the data better, and the attention weight parameters w_k are calculated through a Mean Squared Error (MSE) fit to our data.

To generate data for model fitting, we first developed categories containing stimuli which vary on two or three distinct dimensions. Conducting our experiments on *cognition.run*, we presented all stimuli from training sets defining our categories randomly (with each shown five times) during a training stage. Then the subjects completed a testing stage where every stimuli was shown eight times, and their category responses were captured and totalled.

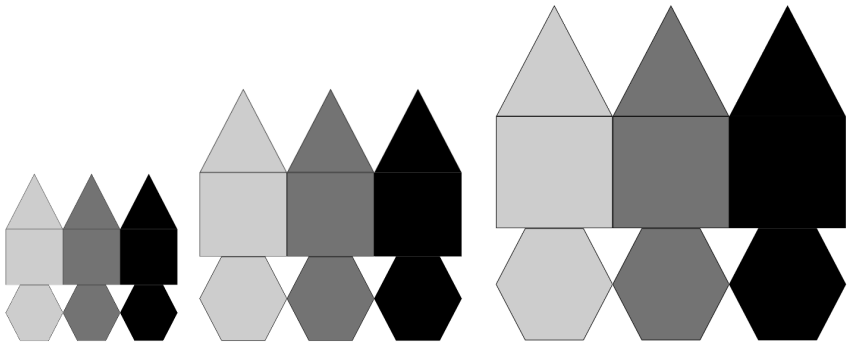

Figure 1: Stimuli varying over three dimensions (# of sides, size, and shade), with three distinct steps in each dimension.

The category training data we generated for two-dimensional and three dimensional cases are as follows, with the motivation behind diagonal and corners being similar, with one or two of the categories in their extreme (two-dimensional vs three-dimensional). The dimensional in two-dimensions acted as a control to make sure our model was acting similarly to the homework (which implemented this category for four steps in each one dimension). In three-dimensions, the interior-exterior category was our own creation, leveraging the ability to distinguish between three dimensions.

Figure 2: Training data for the two-dimensional experiments; each category contains two stimuli.

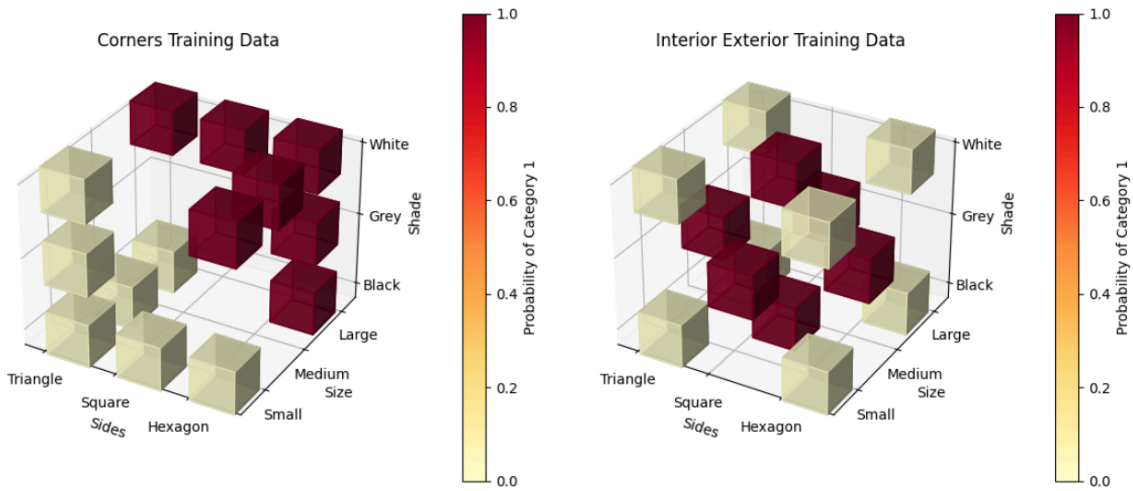

Figure 3: Training data for the three-dimensional experiments. In the corners' data, there are seven stimuli per category, and in the interior-exterior data there are eight stimuli for category two and six for category one.

The one main modification we needed to make for our code came from the computeGCM function, which computes the probability that a stimulus belongs to category one. We also needed to change how the fitParams function worked, as the homework code simplified the model. The computation of weights for each dimension also needed to be added, and the number of steps per dimension changed from four to three. After doing some research on visualization tools in Python, we used Matplotlib to create voxel representations of our model's 3D results and the accumulated data for presentation above.

Results

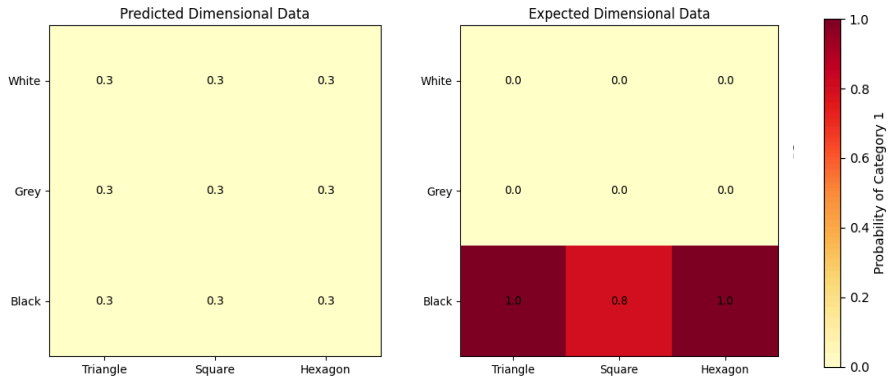

Figure 4: A comparison between the predicted (left) and expected (right) distribution for the 2D Dimensional Condition. The predicted distribution uses the best fit weights and biases that were found to be b=[0.3, 0.7] and w=[0, 1] with an MSE of 0.197.

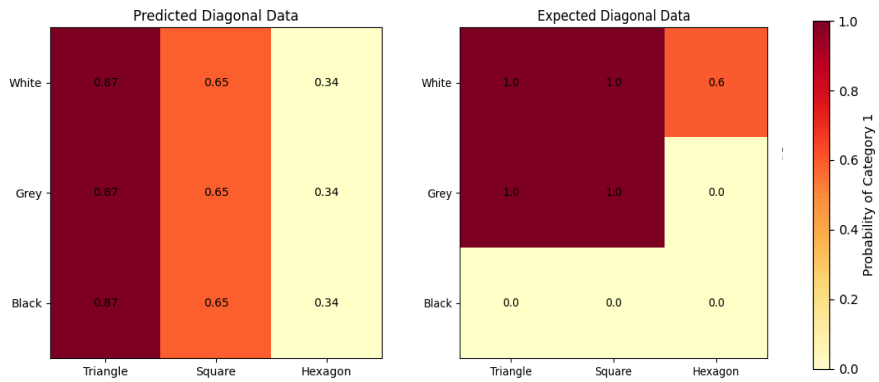

Figure 5: A comparison between the predicted (left) and expected (right) distribution for the 2D Diagonal Condition. The predicted distribution uses the best fit weights and biases that were found to be b=[0.65, 0.35] and w=[0, 1] with an MSE of 0.358.

For the most part, our model ended up performing somewhat awkwardly on the 2D Conditions. By looking at Figures 4 and 5, one can tell that the model wasn't able to capture the expected distribution. For the Dimensional Condition, the model minimized its MSE by completely ignoring the one feature that mattered. For the Diagonal

Condition, the model once again ignored a whole feature. The most likely reason for this is that the subject in each condition had very decisively decided what category each object belonged to–even the objects outside the training set. This is in stark contrast to how the model behaves, where nothing gets 100%, even the training data. We imagine that if we were able to test many more subjects on these conditions then the seemingly arbitrary, yet decisive decisions each subject makes would average out and turn into something closer to what the model predicts. For example, our subject in the Dimensional Condition decided that all grey objects were Category 2. We are not sure if this decision was arbitrary, but we expect that with more subjects many others might instead decide that all grey objects are Category 1.

Although the Corners Condition's MSE was not the best of all the conditions, it by far did the best at capturing the subject's actual responses. As can be seen in Figure 6, for the most part, the subject did what the model predicted. The biggest difference between the model's predictions and the subject's data seemed to come from some confusion the subject had between the small, black hexagon and the large, black hexagon. This is something that the model could not reasonably predict and we would expect to go away with more test subjects. The other noticeable difference between the model and subject was–once again–how decisive the subject was with several objects. Surprisingly though, this was much less apparent in this condition compared to all the others since the subject wasn't totally decisive for all the objects and the model was generally more confident than it was in the Interior Exterior Distribution.

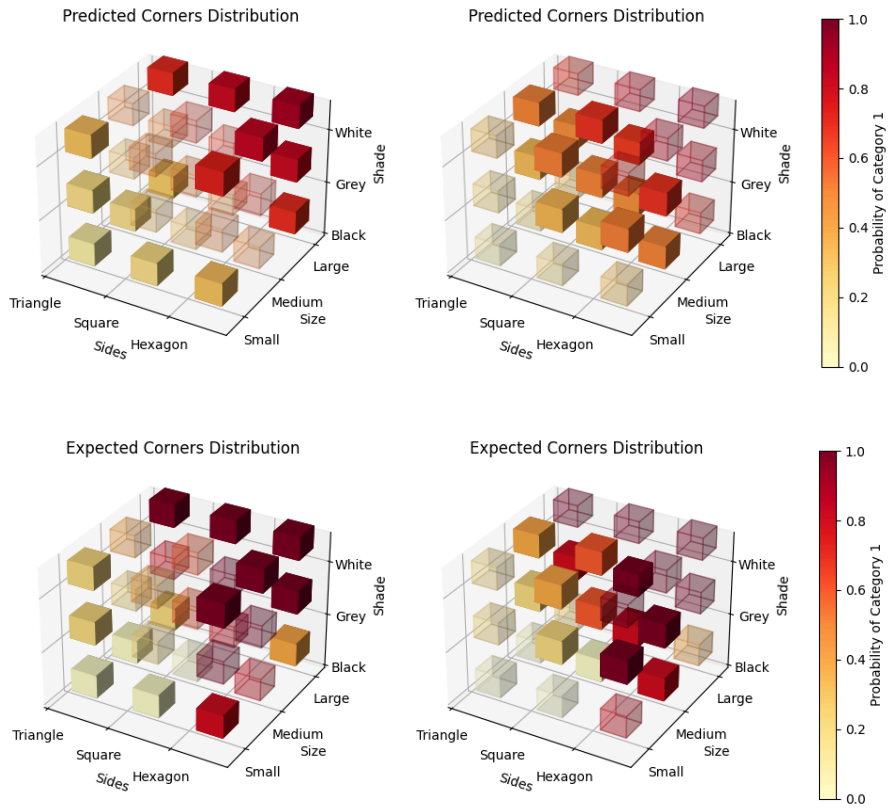

Figure 6: A comparison between the predicted (top) and expected (bottom) distribution for the 3D Corners Condition. The predicted distribution uses the best fit weights and biases that were found to be b=[0.5, 0.5] and w=[0.35, 0.3, 0.35] with an MSE of 0.296.

Finally, for the Interior Exterior Condition, the model was a bit of a let down. Unlike in the Corners Condition, the model here was very unconfident for all of its predictions while the subject was super decisive in picking the category of each object. The one interesting thing of note here is that if you look at the expected Interior-Exterior Distribution in Figure 7, you'll see that the subject decided that the middle of all the vertical edges were Category 2, while the middle of all the horizontal edges are Category 1. The fact that this applies to every single edge makes us think this might not be a coincidence, however, we are unsure of what the reason for this might be.

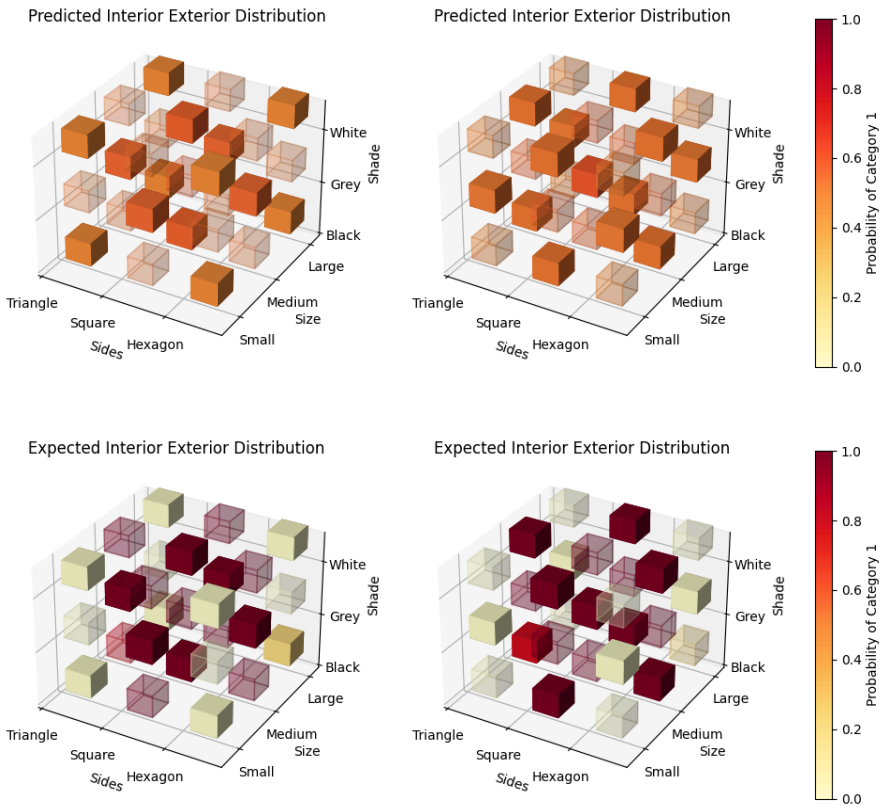

Figure 7: A comparison between the predicted (top) and expected (bottom) distribution for the 3D Interior Exterior Condition. The predicted distribution uses the best fit weights and biases that were found to be b=[0.5, 0.5] and w=[0.35, 0.35, 0.3] with an MSE of 0.341.

Contributions

Both team members worked in-class on research, brainstorming ideas and creating the categories used in the experiments. Henry designed the experimental images and created four cognition.run experiments to be completed by subjects. Dylan drove the codebase change from homework three to incorporate the new model with three categories possible and three dimensions. Dylan also made visualizations of the results of the experiment. Both teammates compiled this report, created the in-class

presentation, and were vocal through the project's lifecycle. We would like to acknowledge Dr. Jern, who helped us to bring this project to fruition and advised us against potential pitfalls.

References

Nosofsky, R. M. (1986). Attention, similarity, and the identification–categorization relationship. Journal of Experimental Psychology: General, 115(1), 39–57. https://doi.org/10.1037/0096-3445.115.1.39